

**SINGAPORE POLYTECHNIC
SCHOOL OF COMPUTING**

COMMON ICT PROGRAMME (Year One)
DIPLOMA IN APPLIED AI AND ANALYTICS (Year One)
DIPLOMA IN CYBERSECURITY AND DIGITAL FORENSICS (Year One)
DIPLOMA IN COMPUTER SCIENCE (Year One)

Module Code	: ST0501
Module Name	: Front-End Web Development
Acad Year/Semester	: 2026/2027 Semester 1
Assignment Title	: CA1 – Static Website
Assignment Type	: Individual
Weightage	: 40%
Deliverables & Deadline (what & when)	: All files used in the web pages. Part 1 - 8 May 2026, Friday, 11:59 pm Part 2 - 29 May 2026, Friday 11:59 pm Digital submission in Bright Space using File naming convention as follows: CCC1AXX_nnnnnnn_yourname.zip Where CCC is your course name e.g. DCS; 1AXX is your class number e.g. 1A01 and nnnnnnn is your admin number e.g. 2312345.
Late submission after the deadline	: 50% of the marks will be deducted for assignments that are received within ONE (1) calendar day after the submission deadline. No marks will be given thereafter. Exceptions to this policy will be given to students with valid LOA on medical or compassionate grounds. Students in such cases will need to inform the lecturer as soon as reasonably possible.

Learning Outcomes

- Critique User Experience and Web Usability by applying C.R.A.P design principles and evaluating factors that contribute to positive web user experience.
- Demonstrate proper HTML page structure by creating clear, well-organized semantic web pages using appropriate HTML tags.
- Apply CSS effectively by using selectors, properties, and box model to create maintainable and visually aesthetic designs.
- Implement website management practices including effective file structures, performance optimization, and validation tools.

Note : GenAI tools may be used, but students must critically assess, refine, and justify AI-generated output.

Assignment Theme: Sustainability

Focus: Choose any sustainability topic that interests you. Examples include:

- **Energy Conservation:** Solar power, energy-efficient appliances, reducing electricity consumption
- **Water Conservation:** Rainwater harvesting, greywater systems, reducing water waste
- **Zero-Waste Lifestyle:** Reducing household waste, reusing materials, composting
- **Sustainable Transportation:** Electric vehicles, cycling, public transport, carpooling
- **Sustainable Food:** Organic farming, local produce, plant-based diets, food waste reduction
- **Green Building:** Eco-friendly construction materials, passive design, green roofs
- **Circular Economy:** Recycling, upcycling, product lifecycle management
- **Renewable Energy:** Wind power, hydroelectric power, biomass energy
- **Sustainable Fashion:** Ethical clothing, textile recycling, slow fashion
- **Urban Sustainability:** Green cities, vertical farming, sustainable urban planning

Website Structure of 3 pages as follows:

1. **Home Page**
 - Introduction to your chosen sustainability topic
 - Relevant statistics or facts about the environmental impact
 - Real-world example(s) of successful implementations or case studies
2. **Personal Action Plan**
 - Outline your personal interest in this sustainability topic
 - Goals you have set or actions you plan to take
 - Reflect on challenges you might face and how you plan to overcome them
 - How others can get involved or make a difference
3. **About Me**
 - Introduction to who you are
 - Your passion for sustainability and environmental issues
 - Any experiences you've had related to your chosen topic
 - Your vision for a more sustainable future

Technical Specifications:

- Complete the project planning template before starting development.
- Create a website of **three** web pages (Include the first landing page of your website using file name: **index.html**).
- Fair use of free stock images/photos (must give attribution on source of images in your comments).
- You are allowed to use content provided by SP (must give attribution on source of images in your comments).
- You are not required to add JavaScript in your assignment.
- You are **not allowed** to use content and designs elements from templates or use frameworks like WordPress, Drupal, Joomla, Wix or any other frameworks.
- Make sure your website can be viewed properly using Google Chrome Browser.
- Your website need not be web-responsive, but website must be viewable with screen resolution of 1366x768 or 1920x1080.
- Web Graphics file size should **not** exceed 800KB (per graphic file).
- All Styling, Layout and Graphic effects should be accessed via external files.
- No fatal errors in validating web pages and CSS styles.
- Add comments in all HTML and CSS files.
- Organization of files (there should be folders to segregate different types of files, e.g. HTML, CSS, and graphics files).
- Apply proper file naming convention
- Consider basic accessibility practices such as meaningful alt text for images and appropriate color contrast.

Deliverables:

CA1	What	When	Where
Part 1	Completed Project Planning template	Week 3 : 8 May 2026, Friday, 11:59 pm	Brightspace
	Project Planning Review	Week 4, first FED lesson	Classroom: Face-to-face
Part 2	1. Website source codes 2. Reflection journal 3. AI prompt logs documentation 4. Academic Integrity Declaration form	Week 6: 29 May 2026 Friday, 11:59 pm	Brightspace
Part 3	Demo cum interview	Week 11: Both FED lessons Week 12: First FED lesson only	Classroom: Face-to-face
	Your tutor will provide a date/time schedule for you to attend the COMPULSORY interview session. Using your laptop, you will be given around 10 - 15 minutes to demo your website, followed by an interview/Q&A session. Students who do NOT turn up for interview without a valid LOA will FAIL CA1 even though all required deliverables are submitted. IMPORTANT: Zip all the relevant files and folders into a single compressed file and name your compressed file using your course name, your class number, your Admin Number and your name: E.g. DCS1A01_2312345_Peter_Lim.zip and submit it via Brightspace.		

Warning:

Plagiarism means passing off as one's own the ideas, works, writings, etc., which belong to another person. In accordance with this definition, you are committing plagiarism if you copy the work of another person and turn it in as your own, even if you would have the permission of that person.

Plagiarism is a serious offence, and if you are found to have committed, aided, and/or abetted the offence of plagiarism, disciplinary action will be taken against you. If you are guilty of plagiarism, you may fail all modules in the semester or even be liable for expulsion.

Marking Scheme:

1	Project Planning & Requirements	10 marks	• Clear sustainability topic selection with rationale • Well-defined content structure for all 3 pages • Appropriate visual design concepts and layout planning
2	Reflection Journal	15 marks	• Critical Analysis of Design Decisions ✓ Thoughtful evaluation of visual design choices ✓ Justification of layout and styling decisions ✓ Analysis of content organisation and structure • Learning Insights and Challenges ✓ Clear reflection on learning process and skill development ✓ Identification of challenges faced and solutions implemented ✓ Personal growth and understanding demonstrated • AI Usage Documentation and Critical Assessment ✓ Comprehensive AI prompt logs with clear documentation ✓ Critical evaluation and refinement of AI-generated outputs ✓ Justification of AI-assisted decisions and human oversight
3	User Experience and Web Usability	15 marks	• Website demonstrates C.R.A.P design principles appropriate for target audience • Navigation is intuitive and information is easy to locate • Content is well-written, accurate, informative, and valuable for users • Website provides good user experience with clear visual hierarchy • Web pages are well-organised and aesthetically pleasing
4	HTML coding	15 marks	• Organize page content with <i>HTML5</i> sectioning elements. • Link to files (using relative paths) within a website and external website. • Include special characters and images. • Error-free - validation of website using https://validator.w3.org.

5	CSS coding	15 marks	• Apply text, font style and colors in CSS. • Use contextual selectors • Work with attribute selectors, margins and padding selectors • Use pseudo-classes and pseudo-elements • Define list styles • Page Layout using CSS (Floating elements, Layout grid or CSS Positioning Styles). • Error-free - validation of website using (https://jigsaw.w3.org/css-validator/)
6	General website management	10 marks	• Each image file size <800KB and source quoted • Viewed properly using Chrome, correct screen resolution, Scripting and Styling accessed via external files • Comments in all HTML and CSS files • Files and folders are well-organized, correct mode of submission • Apply proper file naming convention
7	Interview	20 marks	• Clear demo of website and design decisions • Ability to explain technical implementation • Demonstrate ability to analyse front-end development challenges, evaluate alternative solutions, and articulate critical thinking during the interview

-THE END-